Was magnitude estimation necessary?

A secondary-data reassessment of Bard, Robertson, and Sorace (1996)

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Abstract

This paper reassesses Bard, Robertson, and Sorace's case for magnitude estimation in linguistic acceptability research. It separates two claims that are often run together: acceptability judgments are graded, measurable, and methodologically serious; magnitude estimation has a practical advantage over simpler formal judgment tasks. The first claim is the stronger legacy. The second requires later method-comparison evidence. The paper sets up a constrained, prespecified secondary analysis of Sprouse and colleagues' public method-comparison data, gated on data-reuse permission, uses Langsford et al. as article-level evidence about reliability and response-style risks, and keeps computational benchmarks separate as afterlife evidence rather than evidence about magnitude estimation itself.

Keywords: magnitude estimation; acceptability judgments; grammaticality judgments; secondary data; experimental syntax

1 The problem

Bard, Robertson, and Sorace made a methodological claim that was larger than a single elicitation format. Acceptability judgments, they argued, could be treated as graded measurements rather than as informal binary verdicts (Bard et al., 1996). That move helped make experimental syntax a measurement enterprise: judgments could be collected from naive participants, scaled, compared across groups, and subjected to ordinary statistical tests.

The harder question is whether magnitude estimation was necessary for that advance. Magnitude estimation promised a way around the scale restrictions of traditional judgment tasks, giving participants an open response scale and preserving fine distinctions. But later acceptability work did not settle into one privileged elicitation method. Likert ratings, forced choice, yes/no tasks, and Thurstonian models all became part of the measurement toolkit (Langsford et al., 2018; Sprouse & Almeida, 2017; Sprouse et al., 2013).

This paper separates two claims that are often allowed to travel together. One claim is strong and still plausible: acceptability is graded, measurable, and methodologically serious. The other is

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weaker: magnitude estimation has a practical measurement advantage over simpler formal tasks. The first claim can survive even if the second does not.

The empirical half of this is not untouched. The methodology literature has already compared elicitation formats and largely deflated magnitude estimation's special status. Schütze (2016) reports that most or all of its advertised advantages over Likert tasks have since been refuted, pointing to Weskott and Fanselow (2011) and Sprouse et al. (2013); later work weighed the formats on convergence (Sprouse et al., 2013), statistical power (Sprouse & Almeida, 2017), and reliability and response style (Langsford et al., 2018).

So the bare verdict that magnitude estimation carries no special advantage is not what this paper adds. The contribution is to separate that verdict from the durable measurement claim, to relocate acceptability within a projectibility account, and to route each later dataset to a prespecified estimand, so that any reanalysis of public comparison data, once reuse is permitted, cannot quietly select whichever scale looks best.

The reassessment is therefore not a rejection of Bard et al.'s programme. It is a test of what should be credited to magnitude estimation itself, and what should instead be credited to the broader formalization of acceptability judgment methodology.

2 The Bard target

Reassessing Bard et al. means first stating, from the article rather than its reputation, what they actually claimed. The abstract makes four claims for magnitude estimation: that it can "solve the measurement scale problems which plague conventional techniques", "provide data which make fine distinctions robustly enough to yield statistically significant results of linguistic interest", be "usable in a consistent way by linguistically naive speaker-hearers", and "allow replication across groups of subjects" (Bard et al., 1996, p. 32).

The core of the case is an argument about scale type, not about reliability. The trouble with conventional annotation, they argue, is "the use of the wrong kind of measurement scale" (Bard et al., 1996, p. 32): the asterisk system, like any fixed-point scale, "predetermines the number of distinctions subjects may use" and so censors finer ones (Bard et al., 1996, p. 35). Magnitude estimation is the proposed remedy because it "does not restrict the number of values", giving subjects "complete freedom about which of the infinite set of numbers to use" and yielding interval and ratio data (Bard et al., 1996, p. 41).

This sharpens what is, and is not, Bard et al.'s claim. They do make a method-superiority claim, but a specific one: an open, ratio-yielding response recovers gradient distinctions that fixed-point scales lose by construction. That is a claim about resolution and scale-boundary censoring, and it is testable. The broader reading sometimes met in reception, that magnitude estimation is the privileged formal method in general, goes beyond the article; Bard et al. compare magnitude estimation with informal annotation and fixed-point scales, not with the later inventory of Likert, forced-choice, yes/no, and Thurstonian tasks.

Bard et al. are also more tentative than the reception. They grant that the advantages "are garnered at considerable empirical cost", and they leave open which psychological model underlies acceptability, since "the data currently do not decide between them" (Bard et al., 1996, p. 65). The fair target for reassessment is the resolution claim, not a gold-standard claim the authors did not make.

The distinction matters for design. If the live claim is that an open ratio scale recovers distinctions

Source	Evidence level	Use in this paper	Claim it cannot support
Bard et al. 1996	Published article	Historical target and scale-resolution claim	Direct replication without participant rows
Sprouse 2013/2017	Public row-level comparison data	Constrained secondary method comparison, pending reuse permission	Ungated publication-ready novel results while reuse is pending
Langsford et al. 2018	Published article and supplement	Reliability and response-style synthesis	New participant-level reliability model without raw rows
CoLA, BLiMP, SAD	Benchmark labels and documentation	Afterlife of formal acceptability measurement	Evidence about ME versus other human judgment methods

Table 1: Evidential routing before analysis. ME = magnitude estimation; SAD = Syntactic Acceptability Dataset.

fixed scales censor, then the later method comparisons have to test resolution and scale-boundary effects, not only convergence and reliability. And because no public Bard participant-level rows have been found, this paper cannot replicate the original experiment directly; the reassessment is a conceptual replication and secondary-data test of whether the resolution claim retains independent evidential force.

3 Secondary-data design

The secondary-data design treats the project as a measurement reassessment of Bard, Robertson, and Sorace (Bard et al., 1996). Four evidential routes should stay separate. Published summaries from Bard et al. can source the historical claim, but not a direct replication unless their participant-level data are found.

Later judgment studies can support secondary analysis when they compare elicitation methods on shared or comparable items and provide the rows needed for the intended estimand. Computational benchmarks can show the afterlife of acceptability labels, but they do not bear on magnitude estimation versus Likert ratings. Construct-level discussion can separate acceptability from grammaticality, naturalness, standardness, and correction.

This creates a genre gate. Public row-level data from the Sprouse comparison studies can support a constrained secondary analysis of method performance. The Langsford et al. reliability study is essential evidence for the argument, but unless its raw participant responses are obtained it can only constrain the paper at the article-summary level. The paper should therefore not promise a full response-style or test-retest model across the whole literature.

The Sprouse files also require representation choices before modelling. For Sprouse et al. (2013), the primary forced-choice comparison should use the pair-level sign-test file, because forced choice is a pairwise task and that file already records judgments against expected patterns. The logistic file preserves item-level binary rows, changing the unit of analysis, so it belongs as a named sensitivity specification.

For Sprouse and Almeida (2017), yes/no rows with item IDs `B`, `G`, and `M` should be excluded from item-level comparisons. They are absent from the materials workbook, occur in repeated early order positions for every subject, and do not align with the magnitude-estimation, Likert, or forced-choice item universe. This exclusion is a structural inclusion rule, not an outcome-based trimming decision.

Before any modelling, each source should be assigned to an estimand it can actually answer. The primary estimand is the item-level acceptability signal recoverable across methods. Secondary estimands are method-specific noise or bias and decision agreement for theoretically relevant contrasts.

Participant response style and method-by-item instability are admissible only when the dataset contains the participant-level rows and repeated structure needed to estimate them. This ordering keeps the analysis from selecting whichever scale looks best after the fact, the kind of forking path Gelman and Loken warn about (Gelman & Loken, 2013, 2014).

The substantive comparison is a common-latent-signal null: magnitude estimation, Likert ratings, forced choice, and Thurstonian choice are first modelled as noisy observations of the same item-level acceptability. A magnitude-estimation term becomes interesting only if it improves prediction, reliability, or decision stability enough to matter. Scrambled labels, method-label permutations, and simulations from a no-method-advantage latent model are negative controls for the pipeline, checking whether the analysis manufactures a method winner from degraded structure. They are not substitutes for the substantive common-signal comparison.

4 Scale and reliability

The scale question should be phrased as practical measurement advantage over the common-latent null. Bard et al.'s case for magnitude estimation included scale resolution, fine-grained distinctions, naive-participant use, and replication across groups (Bard et al., 1996, p. 32). The safer target is that source-grounded claim, together with the stronger reading of magnitude estimation that took hold in parts of the later methodology literature. Later method-comparison data give the paper a sharper question: does magnitude estimation recover item differences, participant agreement, or theoretically relevant decisions that simpler tasks lose?

Sprouse et al. (2013) compare formal judgment tasks with informal published judgments, giving evidence about convergence between magnitude estimation, Likert ratings, and forced choice. The public Sprouse rows can support a bounded reanalysis of item-level agreement and decision consequences, subject to reuse permissions. Langsford et al. (2018) add a reliability and response-bias target by comparing Likert, magnitude estimation, forced choice, and Thurstonian models. Without their raw rows, Langsford can support the synthesis of published reliability findings but not a new participant-level response-style or test-retest model.

The reliability analysis inherits the unit-of-analysis charter fixed in the secondary-data design rather than reopening it. The same primary and sensitivity representations carry over, so reliability claims stay traceable to fixed units of analysis instead of to whichever representation later proves most convenient.

The primary reliability analysis should compare methods on prediction, stability, decision consequences, mean separation, and correlation, with priority on consequences for the linguistic contrast. A useful result would show whether magnitude estimation changes the answer for the linguistic contrast under study, reduces Type S or Type M risks (Gelman & Carlin, 2014), or carries costs that offset any added resolution. Sensitivity checks should vary transformations and exclusion rules as named specifications rather than as invisible post hoc choices (Steegen et al., 2016).

5 The contemporary afterlife of acceptability

Contemporary benchmark resources matter because they show what survived from the formal acceptability programme. CoLA turns examples from linguistics publications into binary labels for model evaluation (Warstadt et al., 2019). BLiMP turns expert-designed grammatical contrasts into large sets of minimal pairs (Warstadt et al., 2020). These resources inherit the idea that acceptability-related contrasts can be made portable, scored, and compared across systems.

They also change the evidential object. CoLA's labels come from published linguistic examples rather than from a new sample of naive participant responses. BLiMP's contrasts are generated from expert-crafted grammars and then used to test whether a model prefers the acceptable sentence in a pair. Those are useful evaluation resources, but they are not method-comparison data for magnitude estimation, Likert ratings, forced choice, or yes/no judgments.

The Syntactic Acceptability Dataset is the most useful bridge case because it keeps grammatical status and crowd acceptability status apart (Juzek, 2024). That separation is exactly the level discipline this paper needs. Benchmark labels can show the afterlife of formal acceptability measurement, while participant-response datasets are needed to compare human elicitation methods.

6 Grammaticality and acceptability

The reassessment also needs a disciplined vocabulary. The background view adopted here is that grammar is a system of projectibility profiles (Goodman, 1955): relatively stable patterns that license inferences from forms, meanings, contexts, speakers, and judgments to uses a community or register treats as available. A grammaticality claim is therefore not a report of one response. It states what a form projects to: interpretation, production, correction, expert classification, register fit, cross-speaker stability, and related consequences.

Participant responses under an acceptability prompt are observations of judgments under a task. Expert asterisks and published example labels are classifications made within a theoretical or editorial practice. Benchmark labels are dataset-construction decisions. Each can enter a grammaticality profile, but none is identical to the profile.

This discipline keeps neighbouring constructs apart. Acceptability is one evidential route into a projectibility profile, not the profile itself. Personal production, naturalness, standardness, correction behaviour, and theoretical classification project differently. A sentence can be dispreferred without being ungrammatical, correctable without being unnatural, or standard in one register while unacceptable under another task framing.

Bard et al. helped make one part of this space measurable: graded acceptability judgments from participants (Bard et al., 1996). That is a major contribution to the measurement of one evidential channel. It does not by itself show that an acceptability scale captures every inference licensed by a grammar. Later benchmarks sometimes repackage acceptability-related labels as grammaticality targets (Warstadt et al., 2019, 2020). The Syntactic Acceptability Dataset is valuable here because it explicitly separates literature-derived grammatical status from crowd acceptability status (Juzek, 2024).

That boundary also limits the no-new-data paper. Existing datasets can compare the stability of elicitation methods and clarify how acceptability labels travel. They cannot settle a full projectibility profile for a construction: production, correction, register, interpretation, and community licensing would require data designed for those inferences.

7 Conclusion

The balanced update is straightforward. Bard et al. were right to make acceptability measurable and methodologically serious. Their documented case for magnitude estimation concerned scale resolution, fine distinctions, naive-participant use, and replication across groups (Bard et al., 1996, p. 32). That claim deserves credit as part of the history of experimental syntax.

The method-specific conclusion is more limited. Magnitude estimation has to be compared against a common-latent-signal null, where later formal tasks are treated first as noisy measures of the same item-level acceptability signal. The question is whether magnitude estimation improves prediction, reliability, decision stability, or construct interpretation enough to matter for linguistic work.

The paper's final genre depends on the data gate. If Sprouse reuse permission is available, the project can become a constrained secondary analysis of row-level method-comparison data. If reuse remains unavailable, the project should stay a synthesis and design paper, using Sprouse for verified public structure and Langsford et al. for article-level reliability evidence.

In either genre, the central conclusion is the same. The durable legacy is the formal measurement of graded acceptability as one route into grammar's projectibility profiles. The privileged status of magnitude estimation is a separate claim, and later evidence has to earn it separately.

Data and code availability

No raw participant-level data are committed to this repository. Source and dataset verification are tracked in `notes/source-verification.md`; the analysis charter and structural Sprouse inventory are tracked in `notes/analysis-charter.md` and `notes/dataset-inventory.md`. Scripts currently generate ignored local structural derivatives and readiness manifests, while novel outcome analysis remains gated pending Sprouse data reuse permission or an explicitly verification-only framing.

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TODO: Add final AI-use disclosure with the actual model list used on the project.

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